

Shim's Scheme and Cryptanalysis Implementation

This project implements an identity-based aggregate signature (IDAS) scheme proposed by Shim and a targeted cryptanalysis attack. A total of 2 C code files are developed based on the PBC cryptography library. Shim's IDAS generates 40 private key and 40 individual signatures. The attack generates 39 private key and an aggregate signature which is a summation of 40 individual signatures.

The attack algorithms relies on the Wagner algorithm (a probabilistic birthday search algorithm). To ensure deterministic attack results, we modify the hash generation logic and replace probabilistic Wagner searching with fixed hash summation constraints.

1 Source Code Files

1.1 Original Aggregate Signature Schemes

Shim2.c: Implementation of Shim's identity-based aggregate signature scheme.

1.2 Cryptanalysis Attack Programs

Shim2A.c: The attack on Shim's scheme.

1.3 Wagner Algorithm Utility Codes

Test1.c: Wagner algorithm for 128 hash values with sum = 1 (64-bit)

2 Function Descriptions

2.1 Shim's Schemes(Shim.c)

Each program performs the following operations:

1. Generate 40 private keys
2. Verify 40 partial private keys
3. Generate 40 individual signatures
4. Verify 40 individual signatures separately
5. Aggregate 40 signatures and verify the aggregate signature

2.2 Attack on Shim (Shim2A.c)

- TA only generates private keys for vehicles ($ID_1 \sim ID_{39}$)
- Adversary selects an identity ID_0
- Adversary forge an aggregate signature for $\{ID_i\}_{i=0}^{39}$

3 Key Modification: Wagner Algorithm Replacement

The Shim's scheme is attacked using the Wagner algorithm, which is **probabilistic** and cannot guarantee success in every run.

To achieve deterministic and stable attack results, we modify the hash generation logic:

1. In `Shim2A.c`: Force the sum of hash values to be **1**
2. The original probabilistic Wagner algorithm implementations are preserved in `Test1.c`

4 Important Compatibility Notes

4.1 Differences in Hash Bit Length Design

- Tool code: `Test1.c` adopts 64-bit hash length, which is designed for efficient verification of the Wagner algorithm.
- Attack code: `Shim2A.c` adopts 256-bit hash length, aiming to maintain consistency with the standard parameters in the manuscript.

4.2 Incompatibility Conclusion

Due to the obvious difference in hash bit lengths, `Test1.c` cannot be directly compatible or mixed with `Shim2Ac`.

5 Screenshot Requirements

5.1 Shim's IDAS (Shim2.c)

Include three verification results:

1. Verification of 40 private keys
2. Verification of 40 individual signatures
3. Verification of the final aggregate signature

```
spiderman@spiderman-VMware-Virtual-Platform: ~/Downloads/pbc-0.5.14/include$ gcc -o Shin2 Shim2.c -lpbc -lgmp -lcrypto
spiderman@spiderman-VMware-Virtual-Platform: ~/Downloads/pbc-0.5.14/include$ ./Shin2
0th Key is generated!
1th Key is generated!
2th Key is generated!
3th Key is generated!
4th Key is generated!
5th Key is generated!
6th Key is generated!
7th Key is generated!
8th Key is generated!
9th Key is generated!
10th Key is generated!
11th Key is generated!
12th Key is generated!
```

```

38th Key is generated!
39th Key is generated!
Execution time of Setup algorithm: 0.000027 seconds
Execution time of an extraction algorithm from TA's side: 0.002813 seconds
Execution time of an extraction algorithm from Vehicle's side: 0.007196 seconds
Execution time of a signing algorithm: 0.002227 seconds
The e_left is [33583857544288103748386564209141934344609485496252182229278365688165352838257494683158124059790448371948626466100266630045054589, 283235644502890908451772830908793322295265809849823525330716096060556937944253264587597345852468125528929488268169338515358193869]
The e_right is [33583857544288103748386564209141934344609485496252182229278365688165352838257494683158124059790448371948626466100266630045054589, 283235644502890908451772830908793322295265809849823525330716096060556937944253264587597345852468125528929488268169338515358193869]
]
The aggregate signature is valid!

```

5.2 Attack Programs (Shim2A.c)

The TA generates 39 partial private keys for the adversary in Shim's IDAS, but the adversary still can forge an aggregate signature for 40 vehicles.

```

34th Key is generated!
35th Key is generated!
36th Key is generated!
37th Key is generated!
38th Key is generated!
39th Key is generated!
The e_left is [1420539344361018525425356985748882692224339834055854507283923584977942439750489, 924775693413819233932844248413400921675011674975688244275159284096339938746786681]
The e_right is [1420539344361018525425356985748882692224339834055854507283923584977942439750489, 924775693413819233932844248413400921675011674975688244275159284096339938746786681]
]
The aggregate signature is valid!

```

6 Compilation Notes

All programs depend on the PBC library and GMP library.

6.1 Compilation Command

```

gcc -o Shim2 Shim2.c -lpbc -lgmp -lcrypto
gcc -o Shim2A Shim2A.c -lpbc -lgmp -lcrypto

```

6.2 Execution Command

```

./Shim2
./Shim2A

```